

Dial Tool

Guidebook

beta 2.3 · 4 Oct 2026

A watch-dial design tool for DrawBot on macOS

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Start

Dial Tool is one Python file that runs inside DrawBot, a free drawing app for macOS. It draws a dial as a list of rings, three hands and a date window, in millimetres.



Install

Three steps, one file.

- 1 **DrawBot 3.132** or newer, from drawbot.com.
- 2 **dial.py** from github.com/duztman/dial-tool. Branch main is the released version; dev is the newest work.
- 3 Open dial.py in DrawBot and press **⌘R**. The window opens.



Speed

One optional package.

Knockouts and hollow shapes are cut by combining outlines. DrawBot's own routine is slow; skia-pathops is 2 to 14 times faster.

In DrawBot: Python → Install Python Packages → **skia-pathops** → Go. The Export log then starts with “shape combining: skia-pathops”.



Memory

Nothing to save.

Settings are stored as you work. Pressing **⌘R** again, or quitting DrawBot, keeps them: the last session reopens. Export → Save... writes a named copy; Load... opens one. The nine dials in this guide come as such files.



Unit

mm

Every length is in millimetres, drawn at true size. Radii are measured from the dial's centre.



Angle

0° = 12

Clock angles, growing clockwise. 90° is 3 o'clock, 180° is 6, -90° is 9.



Shape

Fills

Everything exported is a filled outline, never a stroke. A 0.12 mm tick is exactly 0.12 mm in Illustrator.



Order

Top first

The ring list is the stacking order. The top ring is drawn last, in front of the others.



Watch

It runs only inside DrawBot: the window, the fonts and the export all come from DrawBot's own Python.



Watch

The canvas preview needs macOS 14 or newer. On an older system the PDF preview takes over, and the log says why.



Watch

A dial names its fonts as macOS does. On a Mac without the font, that ring falls back to Helvetica Neue.

The window

Settings on the left, the dial on the right, time underneath. The next page shows it whole, redrawn in grey from the tool's own measurements, with the field watch loaded.

①

Sections

Dial, Rings, Hands, Date, Export. One is shown at a time.

②

Ring list

Top ring in front. The checkbox hides a ring; double-click a name to change it. The selected ring is the one the settings edit.

③

Ring tools

Add ring... puts a new ring above the selected one. Duplicate, Delete, move up, move down.

④

The ring's settings

One scrolling list. Rows that don't apply to the ring's kind are hidden, not greyed out.

⑤

Preview

Follows every change. Pinch or ⌘-scroll zooms about the cursor; two fingers pan. With Guides on, the selected ring's edges are orange.

⑥

Time

Drag, or type 10:09:36. Now sets the clock's time; Play runs the hands.

⑦

Guides, Canvas · PDF, Fit

Guides are never exported. Canvas is the fast preview; PDF is DrawBot's own viewer, the reference. Fit shows the whole page.

⑧

Status

Ring, font, diameter; then how long the geometry (build) and the drawing (frame) took.



Number

Slider, field and stepper show one value.

Drag for a rough value, type for an exact one, step for the last hundredth. Typing past a slider's end extends the slider.



Typing

Fields do arithmetic.

1.2	sets 1.2
+0.5	adds 0.5
x0.5	halves
/3	a third



Choice

Popups, switches, checkboxes.

A choice can change which rows are shown: pick dot and Length disappears.

Dial Tool · beta 2.3

Dial Rings Hands Date Export

Ring	Kind
☒ Hours	numerals
☒ 24 hours	numerals
☒ Hour marks	markers
☒ Minutes	ticks

Add ring... Duplicate Delete Up Down

Kind ticks Colour Outer radius 14.40

PositionsCount 60Start ° 0.0Span ° 360Every 1Offset 0Skip those instead

Give way to rings aboveSkip where markers aboveKnock out nothing

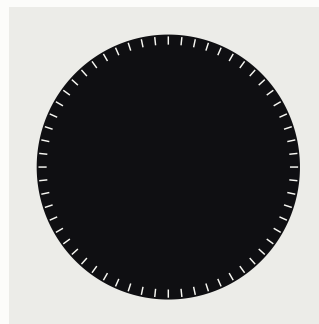
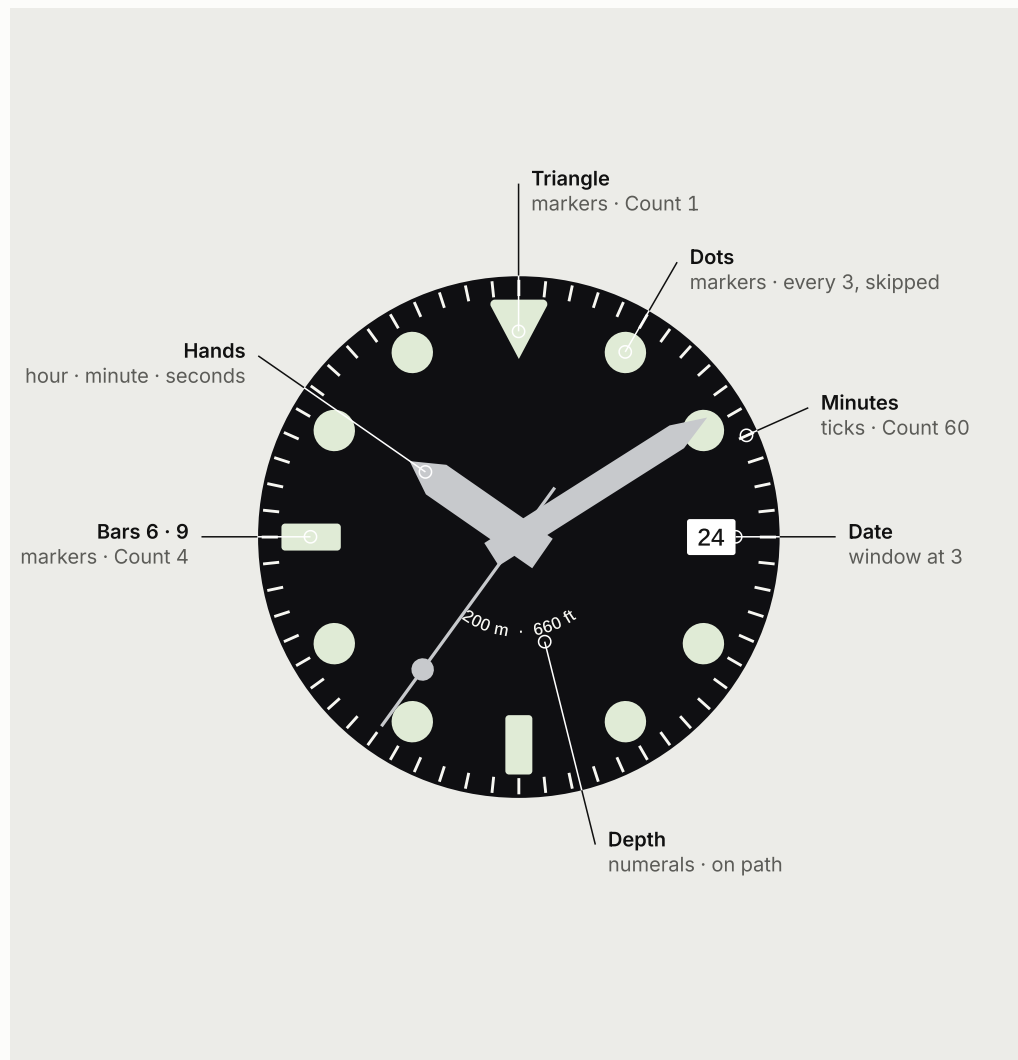
LookShape bar wedge dot fileLength 1.00Width · Ø 0.16Round outer 0.00Round inner 0.00

Minutes · Ø 30.0 mm · canvas · build 0 ms · frame 1 ms

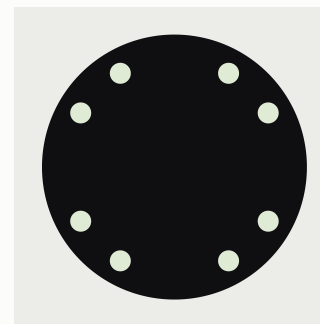
Time 10:09:36 Guides Canvas PDF Now Play Fit

Rings

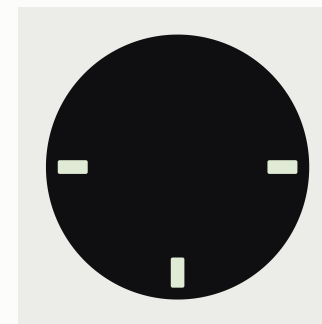
A ring repeats one thing round the centre, at its positions. A dial is a short list of rings, stacked.



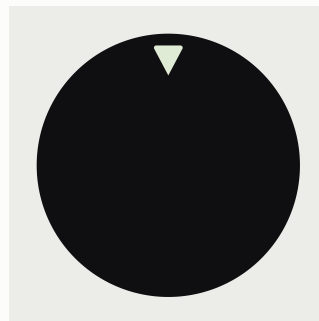
Minutes
ticks · Count 60



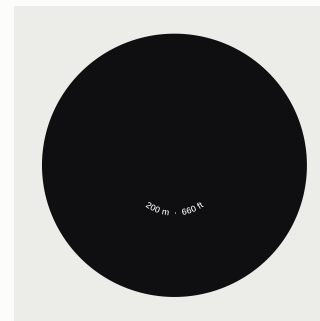
Dots
markers · Every 3, skip those instead



Bars 6 · 9
markers · Count 4 · 12 o'clock: none



Triangle
markers · Count 1 · wedge



Depth
numerals · Custom · on path



The dial
with hands and the date



The idea

One dial, five rings.

The diver of page 16, taken apart. Nothing is special-cased: the triangle is a markers ring with a count of one, the depth rating a numerals ring with one label.



Kind

Four kinds.

Ticks and markers repeat a shape, numerals a label; a band is colour between two radii. A ring's kind can change at any time: it carries every setting.

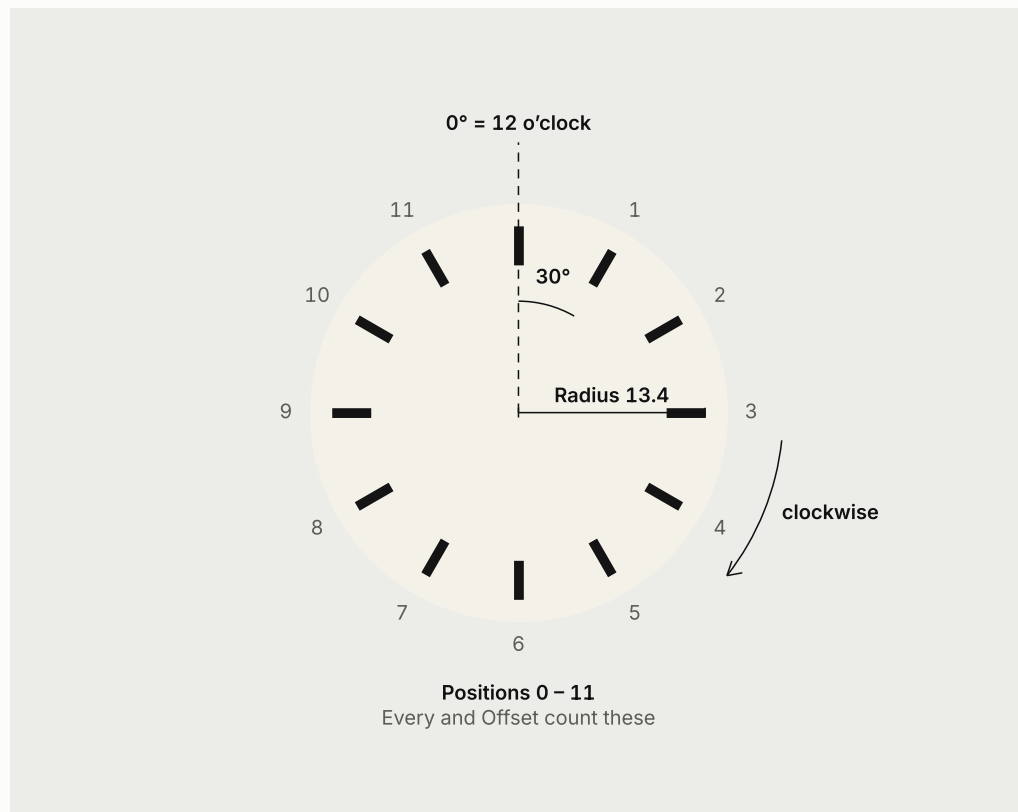


Watch

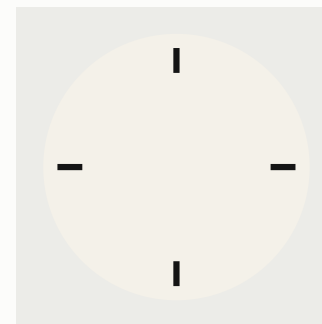
Order works twice. It is the stacking order, and a ring only gives way to rings **above** it: to make ticks stop for markers, the markers go higher in the list.

Positions

Where a ring puts things. Six numbers describe almost any track, scale or set of hours.



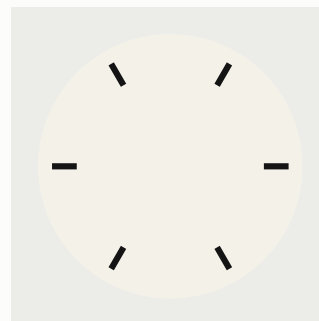
Count 12
Start 0° · Span 360°



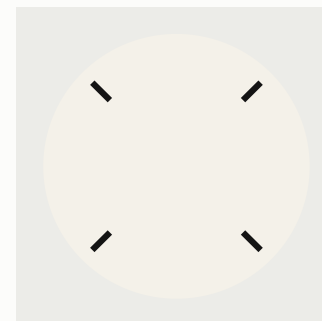
Every 3
the cardinals



Every 3 · skip those instead
the other eight



Every 2 · Offset 1
the odd hours



Count 4 · Start 45°
a quarter turn off



Count 10 · Start -135° · Span 270°
an arc: both ends marked

The arc, sixth picture

Count

Start °

Span °

The cardinals, second picture

Every

Offset

☐ Skip those instead



Watch

Radii are absolute. Changing the dial's Diameter moves no ring and no hand: set the diameter first, then each Radius, as you would for a real movement.



Count · Start · Span

An arc marks both ends.

Span 360° spaces evenly. Less puts the first and last on the ends: Count 10 over 270° runs 0 to 9.



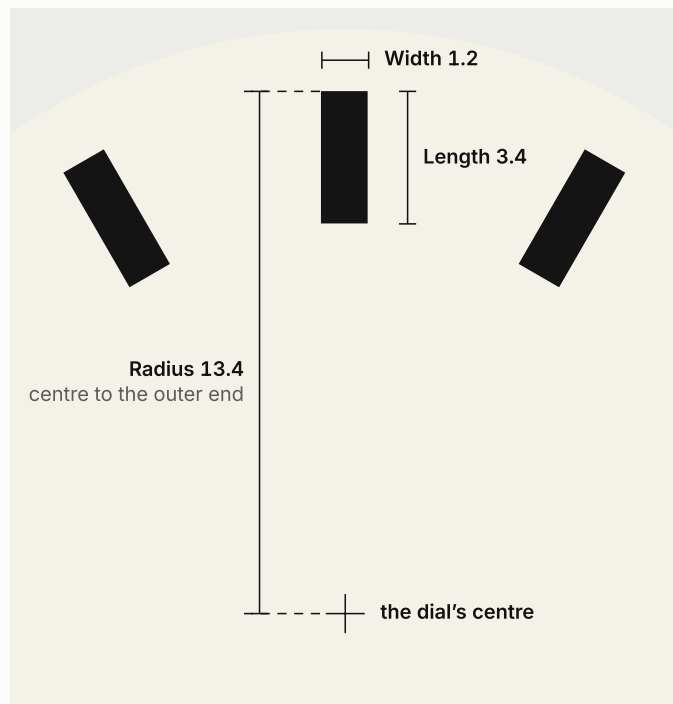
Every · Offset

Use some of them.

Every 3 keeps the cardinals; Skip those instead keeps the other eight.

Marks

Ticks and markers share one set of shapes. A markers ring adds a first position that can differ from the rest.



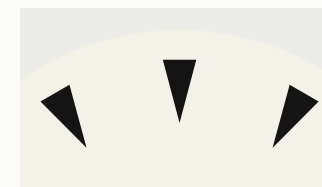
bar
Length 3.4 · Width 1.2



bar, fine
Count 60 · Length 1.6 · Width 0.12



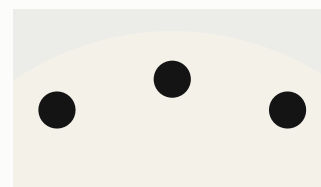
wedge
Inner width × 0.35



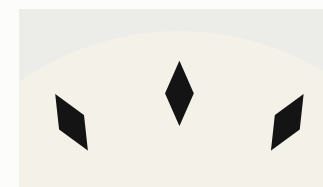
wedge
Inner width × 0



wedge
Inner width × 1.8



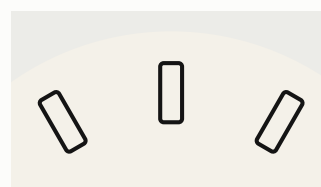
dot
Ø 2.0



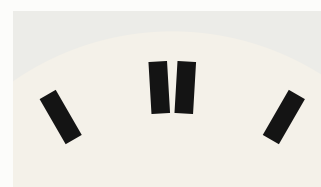
file
your own SVG · PDF · AI



rounded
Round outer 0.6 · Round inner 0.2



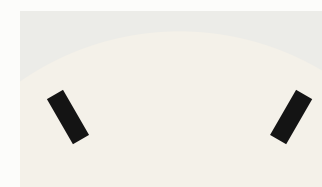
hollow
Hollow wall 0.22



12: double
Gap 0.4



12: triangle
Scale × 1.2



12: none

The mark above

Shape ☒ bar ☐ wedge ☐ dot ☐ file

Outer radius

Length

Width · Ø



Watch

A fillet can't exceed half the side it rounds, and a Hollow wall of half the width or more leaves the mark solid. Neither warns: the shape just stops changing.



Wedge and corners

One number tapers; two round.

Inner width × sets the inner end as a share of the outer: 0 is a triangle, 1 a bar, more than 1 widens inward. Round outer and Round inner are true fillets, in mm.



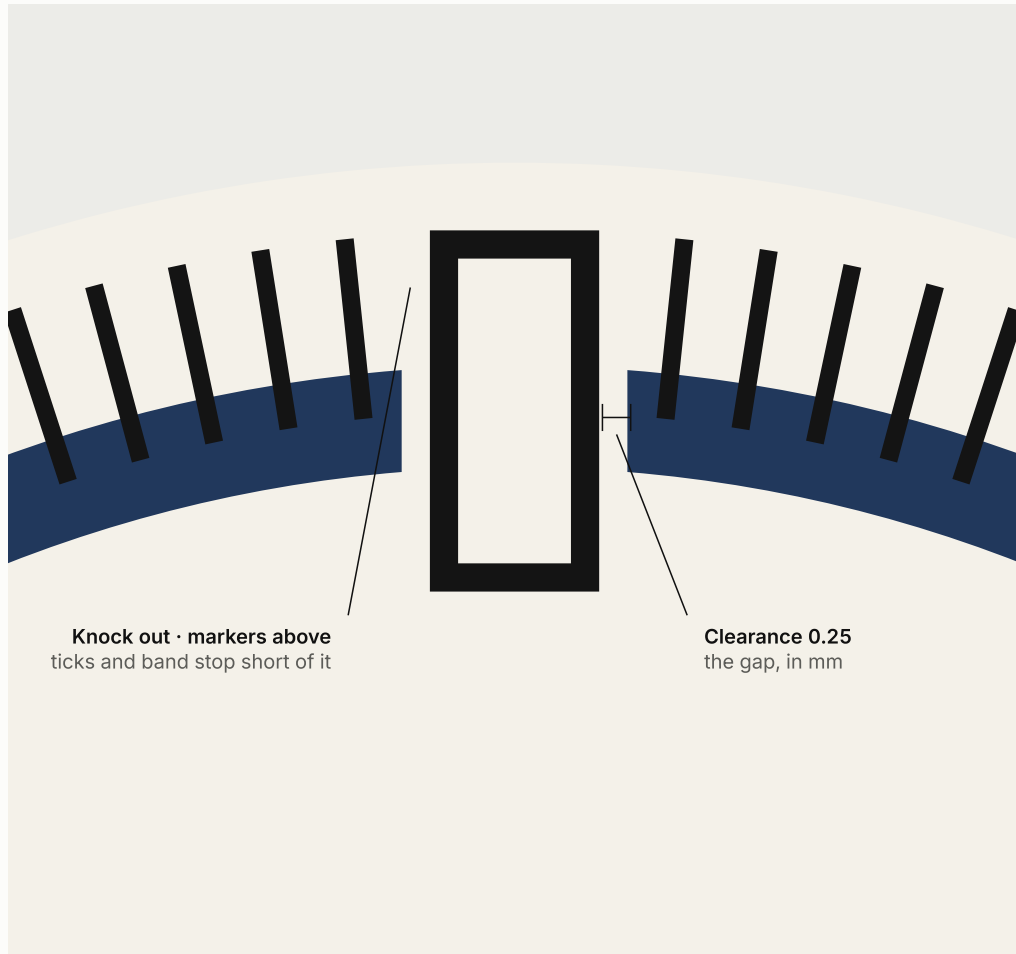
12 o'clock

The first position can differ.

A markers ring draws its first position the same, double (with a Gap), as a triangle, or not at all. Scale × sizes it. Ticks have no such position.

Give way

How a ring reacts to the rings above it. One mechanism covers every pair of rings.



The ticks above

Skip where

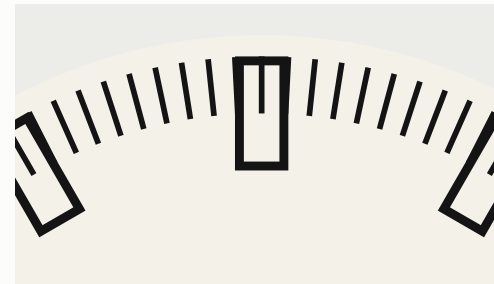
Knock out

Clearance

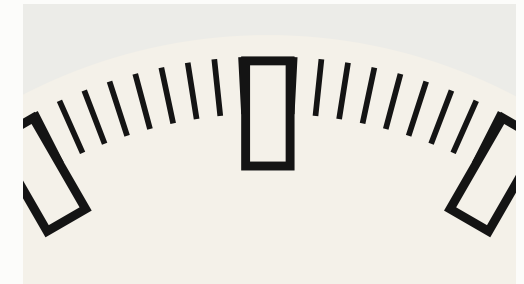


Watch

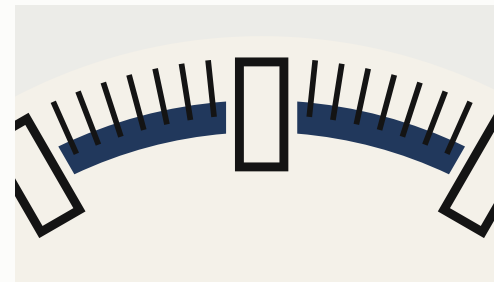
Skip compares angles to a hundredth of a degree. Two rings skip each other only where their positions truly coincide: same Start, counts that divide.



Skip where: nothing · Knock out: nothing



markers above



markers above · Clearance 0.25



markers & numerals above



Skip where

Leave out what is taken.

The ring drops its own positions wherever the chosen rings above already have a mark: minute ticks that stop for the hour markers.



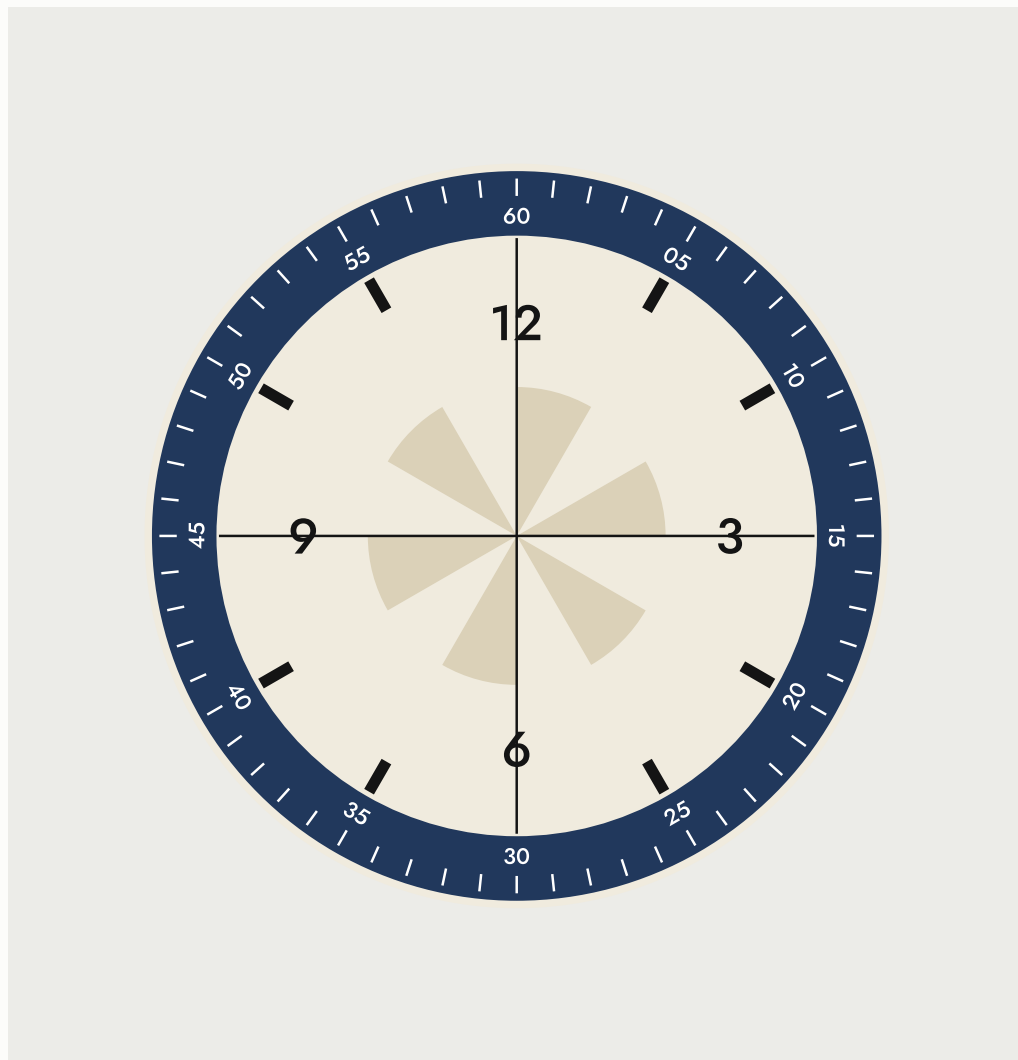
Knock out

Cut a gap round it.

The ring is cut back by Clearance mm round the shapes above: markers, markers and numerals, or everything.

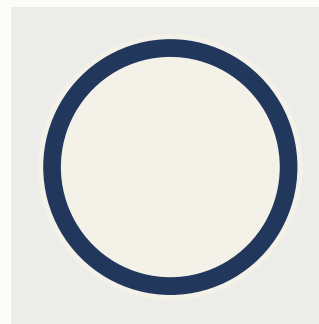
Bands

Flat colour between Radius and Radius – Band width: rings, rails, zones and sectors.



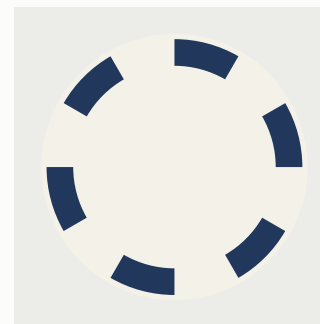
Watch

Colour is flat, one per ring. A chapter ring with white print is three rings: the band, and white ticks and numerals above it — as in this sector dial, page 22.



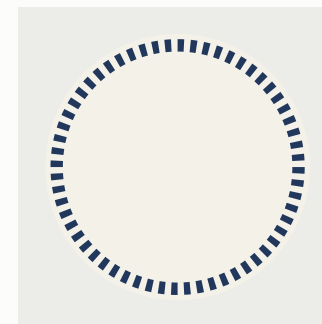
ring

Count 1 · Band width 2.0



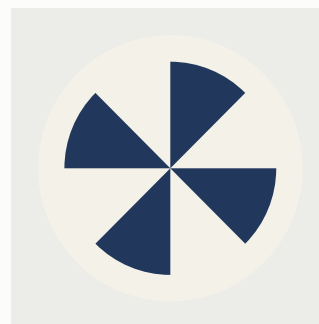
sectors

Count 12 · Every 2



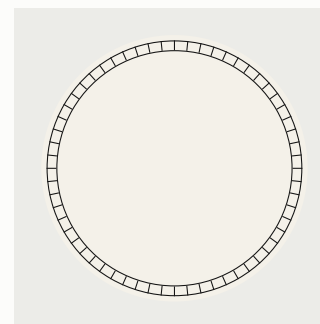
fill

Count 60 · Fill 50 %



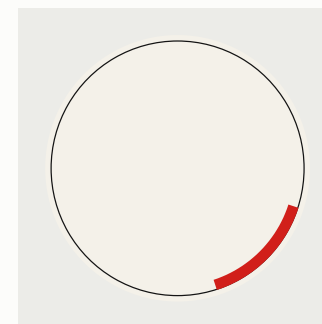
pies

Band width = radius



rails

two bands, Band width 0.12



arc

Start 108° · Span 54°



Segments

Positions cut a band.

Count 1 is a full ring. More positions cut it into segments that tile the span; Fill % is how much of its step each one covers.

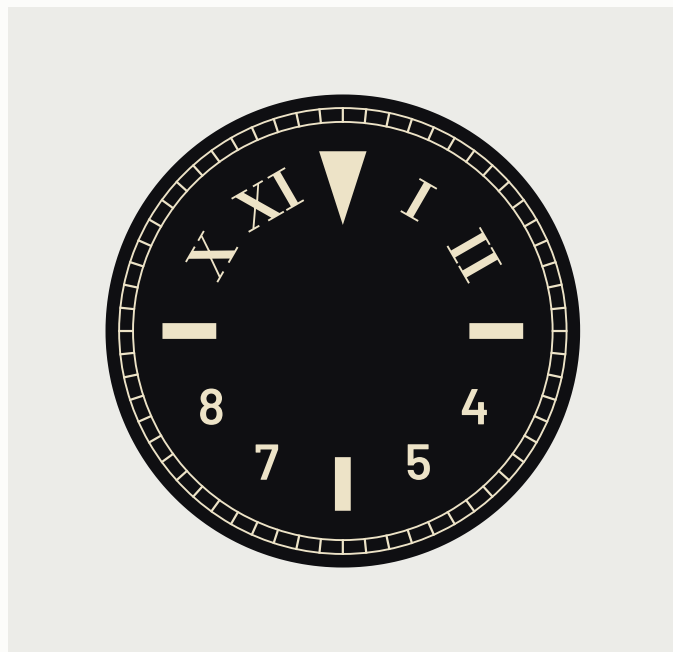


The sectors, second picture

Outer radius	14.40	↑	↓
Band width	3.00	↑	↓
Count	12	↑	↓
Every	2	↑	↓

Numerals

One label per position: from a built-in set, a count, or a list of your own.



Arabic	Roman · III	Roman · IV	Eastern Arabic
Chinese	Words on path	Numbers First 5 · Step 5 · Digits 2 · Start 30°	Custom N, NNE, , , E, ...
upright always level	radial turns with the circle	radial, auto-flip never upside down	on path letters follow the circle

12

Numbers

Labels that count.

Labels
Numbers

First number
5

Step
5

Digits
2



Watch

A Custom list follows positions, not hours. With Start 30° its first entry sits at 1 o'clock.



Placement

Four ways to sit.

On path sets each letter on the circle with the font's own spacing; below 3 and 9 o'clock it flips to read left to right.



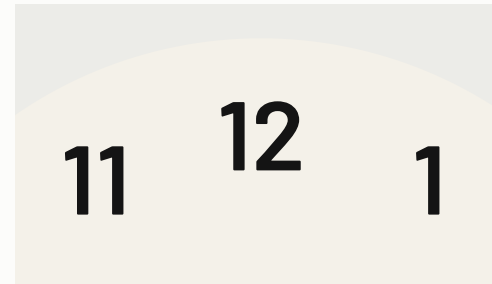
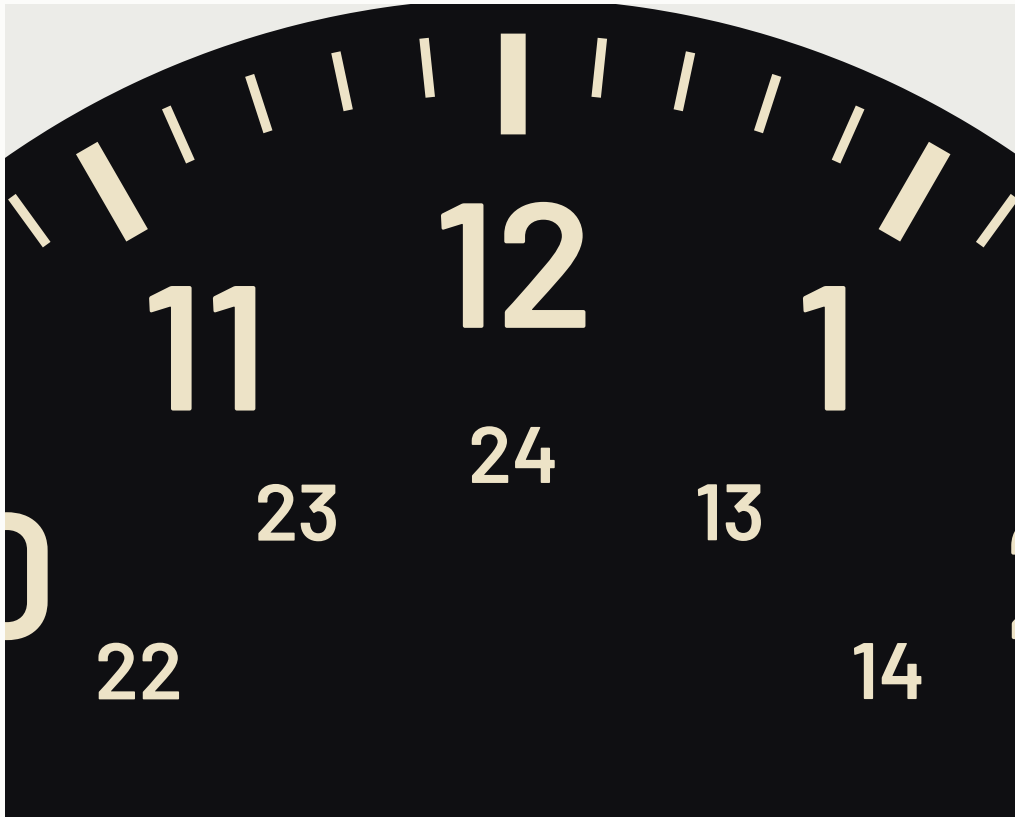
Custom

A list, comma by comma.

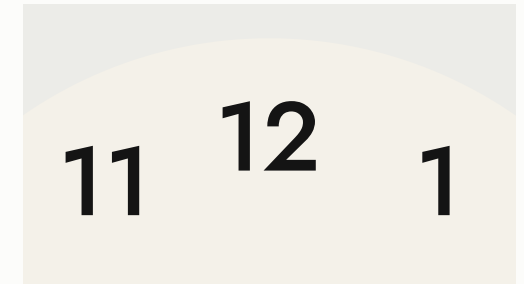
Entries go to the positions in order. An empty entry leaves its position empty: the California dial, page 19, is two such lists.

Type and nudge

Each numerals ring has its own type, from any font installed on the Mac. Each numeral can then be corrected by eye.



DIN Alternate Bold



Futura Medium



Helvetica Neue Medium



Bodoni 72 Book



Nudge

One numeral at a time.

Radius ±	in or out, mm
Angle ±°	round the dial
Rotate °	about its centre
Size ×	larger, smaller



Watch

Two-digit numerals look heavier than single ones. Size × 0.92 on 10, 11 and 12 evens them; Radius −0.2 pulls a wide 12 inward. Nudges are stored per position: change Count and they stay with the position, not the label.



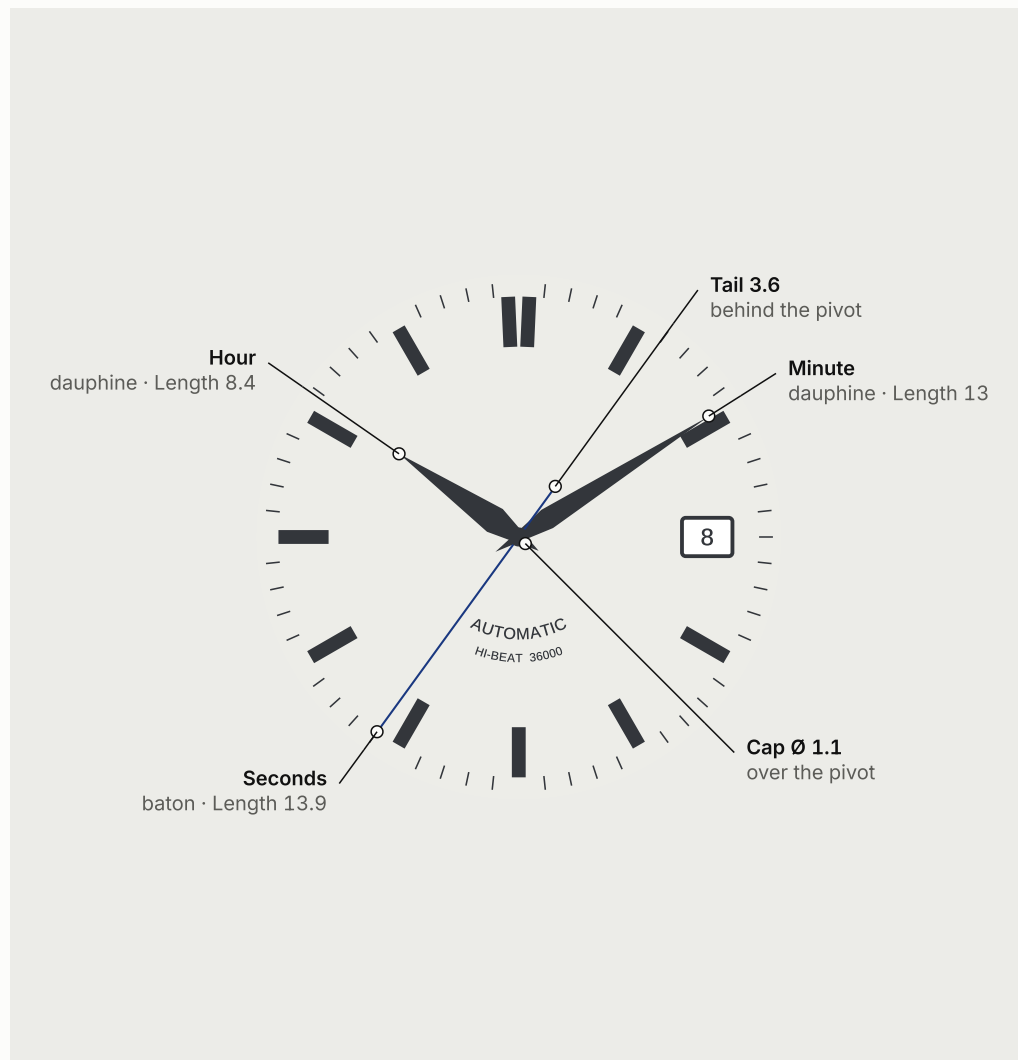
Type

What reaches the outlines.

Family · Style	read from macOS; there is no fonts folder
Instance · Axes	a variable font's named instances, and a slider per axis
Features	OpenType features under the designer's own names
Language · Tracking	local forms, such as Turkish İ i · 1/1000 em, as in Illustrator

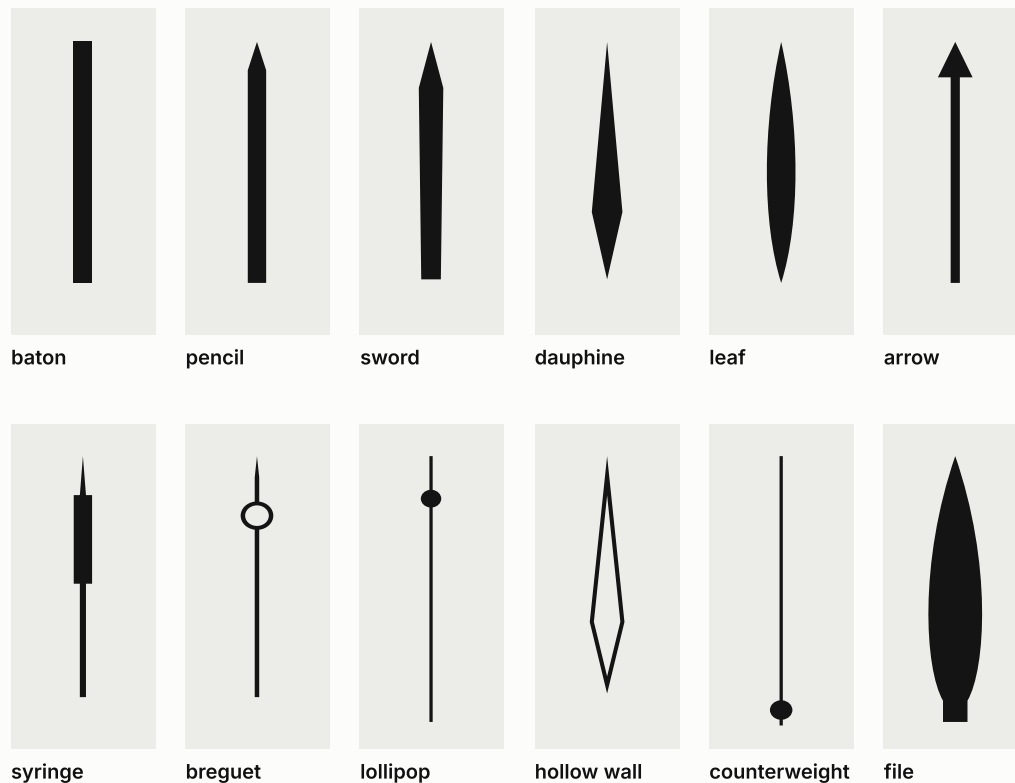
Hands

Hour, minute and seconds each have their own shape, size and colour. Lengths are millimetres from the pivot, as movements specify them.



Watch

Lengths don't follow the dial. A 13 mm minute hand on a 24 mm dial overhangs it by a millimetre; nothing stops you.



The diver's minute hand

Shape	pencil	
Length	<input type="range"/>	12.40
Width	<input type="range"/>	1.50
Tail	<input type="range"/>	1.80
Tip length	<input type="range"/>	2.00



Motion

Four beats.

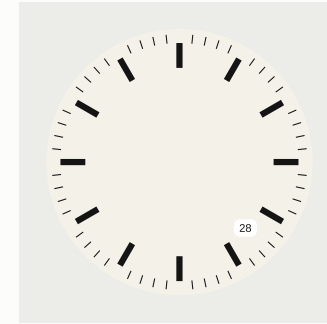
sweep	smooth
quartz	1 step a second
6 · 8 beats/s	21,600 · 28,800 vph

Date window

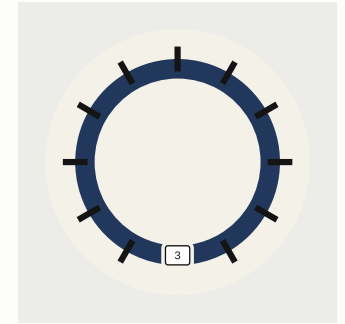
A hole in the plate that shows the date disc, with an optional printed frame. Print keeps its distance from it.



at 3
Radius 10.5 · 2.6 × 2.0 · Frame 0
0.15



at 4:30
Corner 0.6 · Frame 0



at 6
Print clearance 0.4 cuts the band 0.15

17

The window above

Position

Radius

Width

Height

Corner

Frame

Print clearance



Watch

The frame and the day share one Print colour, and the disc is always white. On a dark plate a light frame would mean a light day on white: set Frame to 0 there, as the diver does.



Clearance

Print keeps away.

Every ring is cut back by the frame plus Print clearance. Markers and numerals leave the date's position empty altogether.



Type

The top numerals ring's font.

Shows is the day drawn on the disc; Text % of height sizes it.



Production

A cut line.

In a production mask the aperture is an outline to cut or drill to.



Position

Five places.

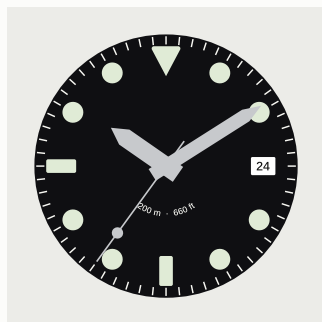
3, 4:30, 6, 9 or 12; Radius sets how far out.

Classics

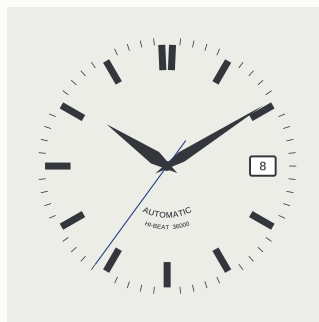
Nine dials drawn after well-known types, by eye: approximations, not measured copies. Each page prints the settings that drew the picture.



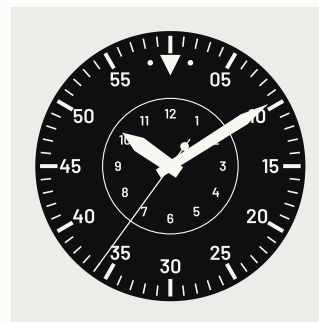
Field watch
page 15



Diver
page 16



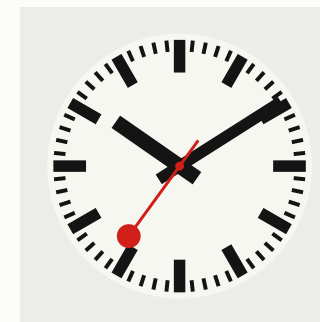
Dress watch
page 17



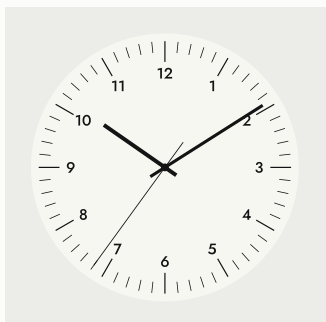
Pilot, type B
page 18



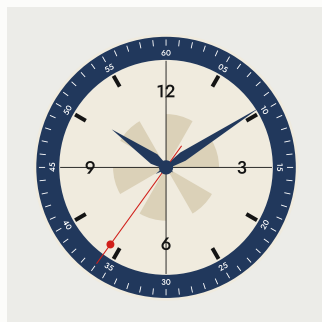
California
page 19



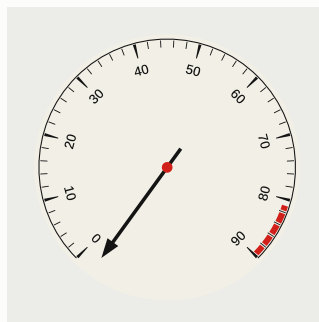
Station clock
page 20



Bauhaus
page 21



Sector dial
page 22



Gauge
page 23



Presets

Load them.

Each dial is a settings file in *docs/guide/presets*. Export → Settings → Load... opens it, ready to change.



Reading a page

Rings top first.

Only what differs from a newly added ring is printed; everything else is at its starting value.



Watch

Fonts are named as on macOS: DIN Alternate, Futura, Helvetica Neue, Bodoni 72. These pages were rendered off-Mac with similar open fonts, so numerals differ slightly from what the presets draw on a Mac.



Watch

The presets have not been opened in the tool on a Mac yet. If one looks different from its page, the page is the intention.



12.1

Field watch

After the A-11 of the 1940s and its descendants: 12 and 24 hours, a plain minute track, nothing else.



Look for

The 24-hour ring is a second numerals ring with a Custom list. The minute ticks **skip** where the hour marks sit, so the two never overprint.

Rings top first, then hands. Only what differs from a new ring.

Hours numerals	Radius 10.5 · Size 3 · Font DIN Alternate Bold ●
24 hours numerals	Radius 7.3 · Labels Custom · Custom 24, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 · Size 1.35 · Font DIN Alternate Bold ●
Hour marks markers	Radius 14.4 · Length 1.7 · Width 0.42 ●
Minutes ticks	Radius 14.4 · Count 60 · Length 1 · Width 0.16 · Skip where markers above ●
Hour hand	Shape sword · Length 7.6 · Width 1.5 · Tail 1.6 · Tip length 2.4 ●
Minute hand	Shape sword · Length 12.6 · Width 1.2 · Tail 1.8 · Tip length 3 ●
Seconds hand	Shape baton · Length 13.2 · Width 0.14 · Tail 3.4 ●
Cap	Ø 1.5 ●
Preset	<i>presets/field.json</i> · Dial 30 mm ●



12.2

Diver

After the 1950s dive watch: fat dots, bars at 6 and 9, a triangle at 12, the date at 3.

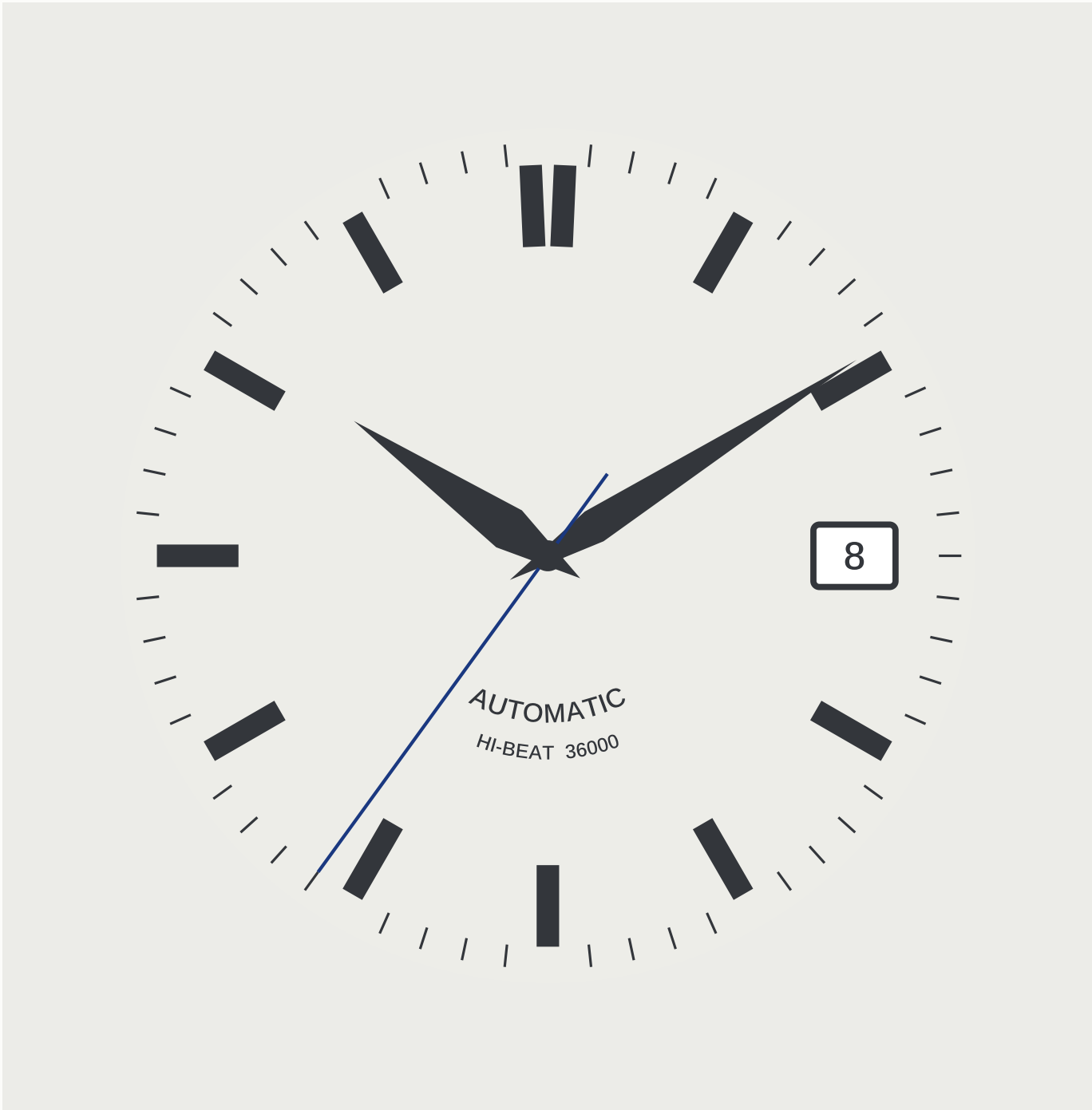


Look for

Three markers rings share the hours: dots on **Every 3**, **skip those instead**; bars on Count 4 with the first position set to none; one triangle. The date takes 3 by itself.

Rings top first, then hands. Only what differs from a new ring.

Triangle markers	Radius 13.2 · Count 1 · Shape wedge · Length 3.3 · Width 3.5 · Inner width × 0 · Round outer 0.25 ○
Bars 6 · 9 markers	Radius 13.2 · Count 4 · Length 3.3 · Width 1.5 · Round outer 0.2 · Round inner 0.2 · 12 o'clock none ○
Dots markers	Radius 13 · Every 3 · Skip those instead · Shape dot · Ø 2.3 ○
Depth numerals	Radius 5.2 · Count 1 · Start 180° · Labels Custom · Custom 200 m · 660 ft · Placement on path · Size 0.95 · Font Helvetica Neue Medium ○
Minutes ticks	Radius 14.3 · Count 60 · Length 0.9 · Width 0.16 ○
Date window	Position 3 · Radius 10.7 · Width 2.7 · Height 2 · Corner 0.15 · Frame 0 · Print clearance 0.3 · Shows 24 · Text 68 %
Hour hand	Shape pencil · Length 7.4 · Width 2 · Tail 1.6 · Tip length 1.8 ○
Minute hand	Shape pencil · Length 12.4 · Width 1.5 · Tail 1.8 · Tip length 2 ○
Seconds hand	Shape lollipop · Length 13 · Width 0.18 · Tail 3.4 · Head · ring · disc 1.25 · at 70 % ○
Cap	Ø 1.6 ○
Preset	<i>presets/diver.json</i> · Dial 29 mm ●



12.3

Dress watch

After the Japanese dress watches of the 1960s: plain bars, a double bar at 12, dauphine hands, a framed date.



Look for
The double bar is the markers ring's 12 o'clock style. Both lines of text are numerals rings with Count 1, set on path at 180°.

Rings top first, then hands. Only what differs from a new ring.

Hours markers	Radius 13.9 · Length 2.9 · Width 0.8 · 12 o'clock double · Gap 0.3 ●
Minutes ticks	Radius 14.7 · Count 60 · Length 0.8 · Width 0.09 · Skip where markers above ●
Line 1 numerals	Radius 5.6 · Count 1 · Start 180° · Labels Custom · Custom AUTOMATIC · Placement on path · Size 0.95 · Font Helvetica Neue Medium ●
Line 2 numerals	Radius 7 · Count 1 · Start 180° · Labels Custom · Custom HI-BEAT 36000 · Placement on path · Size 0.7 · Font Helvetica Neue Medium ●
Date window	Position 3 · Radius 10.9 · Width 2.7 · Height 2 · Corner 0.1 · Frame 0.22 · Print clearance 0.25 · Shows 8 · Text 70 %
Hour hand	Shape dauphine · Length 8.4 · Width 1.6 · Tail 1.4 · at 20 % ●
Minute hand	Shape dauphine · Length 13 · Width 1.25 · Tail 1.6 · at 15 % ●
Seconds hand	Shape baton · Length 13.9 · Width 0.12 · Tail 3.6 ●
Cap	Ø 1.1 ●
Preset	presets/dress.json · Dial 30.4 mm ○



12.4

Pilot, type B

After the 1940s observer's watch: minutes outside, hours on an inner ring, a triangle and two dots at 12.



Look for

The minutes are **Numbers**: first 5, step 5, two digits, over a 300° span that leaves 12 free for the triangle. The two dots are a ticks ring with Count 2 on a 22° span.

Rings top first, then hands. Only what differs from a new ring.

Minutes numerals	Radius 12.6 · Count 11 · Start 30° · Span 300° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Size 2.5 · Font DIN Alternate Bold ○
Triangle markers	Radius 14 · Count 1 · Shape wedge · Length 2.6 · Width 2.6 · Inner width × 0 ○
Two dots ticks	Radius 13.6 · Count 2 · Start -11° · Span 22° · Shape dot · Ø 0.7 ○
Five minutes markers	Radius 16.4 · Length 2.2 · Width 0.5 ○
Minute track ticks	Radius 16.4 · Count 60 · Length 1.3 · Width 0.2 · Skip where markers above ○
Hours numerals	Radius 6.6 · Size 1.7 · Font DIN Alternate Bold ○
Hour circle band	Radius 8.6 · Band width 0.14 ○
Hour hand	Shape sword · Length 7.4 · Width 2 · Tail 1.4 · Tip length 2.6 ○
Minute hand	Shape sword · Length 14.6 · Width 1.5 · Tail 1.8 · Tip length 3.6 ○
Seconds hand	Shape baton · Length 15.4 · Width 0.16 · Tail 4.2 · Counterweight Ø 1.1 ○
Cap	Ø 1.7 ○
Preset	<i>presets/pilot.json</i> · Dial 34 mm ●



12.5

California

After the 1930s–40s dial: Roman on top, Arabic below, bars at 3 · 6 · 9, a triangle at 12, a railroad track.



Look for

Two numerals rings on one radius, each with a **Custom** list that leaves the other's positions empty — so Roman and Arabic can use different fonts and placements.

Rings top first, then hands. Only what differs from a new ring.

Roman numerals	Radius 9.6 · Labels Custom · Custom , <i>I, II, , , , , , , X, XI</i> · Placement radial · Size 3.3 · Font Bodoni 72 Book
Arabic numerals	Radius 9.6 · Labels Custom · Custom , , , , , <i>4, 5, , 7, 8, ,</i> · Placement upright · Size 3.2 · Font DIN Alternate Bold
Bars + triangle markers	Radius 11.4 · Count 4 · Length 3.4 · Width 1 · 12 o'clock triangle · Scale × 1.25
Minutes ticks	Radius 14.2 · Count 60 · Length 1 · Width 0.14
Outer rail band	Radius 14.2 · Band width 0.12
Inner rail band	Radius 13.32 · Band width 0.12
Hour hand	Shape pencil · Length 7.2 · Width 1.5 · Tail 1.4 · Tip length 1.6
Minute hand	Shape pencil · Length 11.8 · Width 1.2 · Tail 1.6 · Tip length 1.8
Seconds hand	Shape baton · Length 12.6 · Width 0.13 · Tail 3.2 · Counterweight Ø 0.9
Cap	Ø 1.4
Preset	<i>presets/california.json</i> · Dial 30 mm



12.6

Station clock

After the Swiss railway clock of 1944:
heavy bars, and a red seconds hand like a
dispatcher's baton.

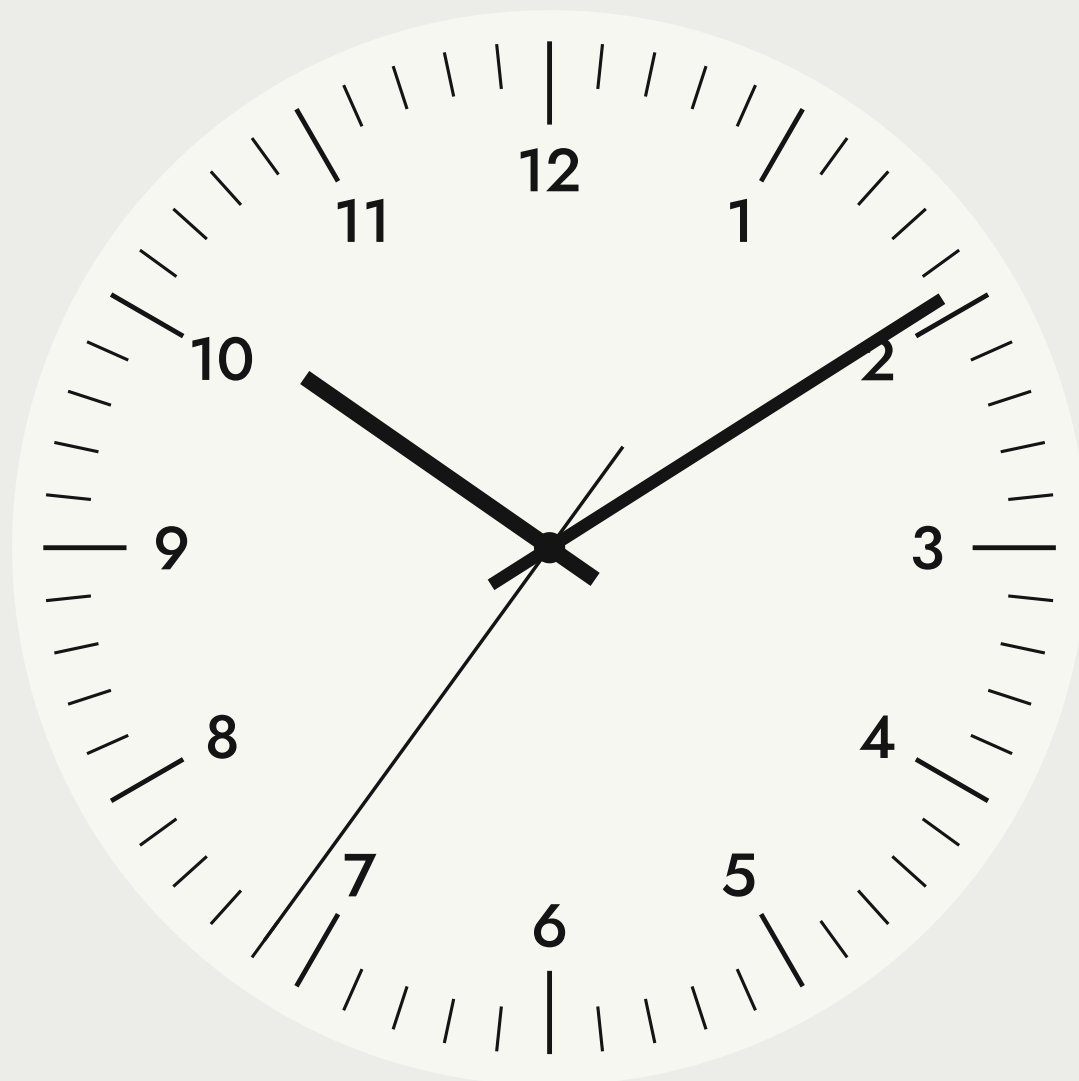


Look for

Two rings. Everything is in the proportions: bars 3.7×1.3 , ticks 1.3×0.45 , and a lollipop with its disc at 100 % of the length.

Rings top first, then hands. Only what differs from a new ring.

Hours markers	Radius 14.3 · Length 3.7 · Width 1.3 ●
Minutes ticks	Radius 14.3 · Count 60 · Length 1.3 · Width 0.45 · Skip where markers above ●
Hour hand	Shape baton · Length 8.8 · Width 1.7 · Tail 2.4 ●
Minute hand	Shape baton · Length 13.6 · Width 1.3 · Tail 2.8 ●
Seconds hand	Shape lollipop · Length 9.8 · Width 0.34 · Tail 3.6 · Head · ring · disc 2.7 · at 100 % ●
Cap	Ø 1 ●
Preset	<i>presets/railway.json</i> · Dial 30 mm ○



12.7

Bauhaus

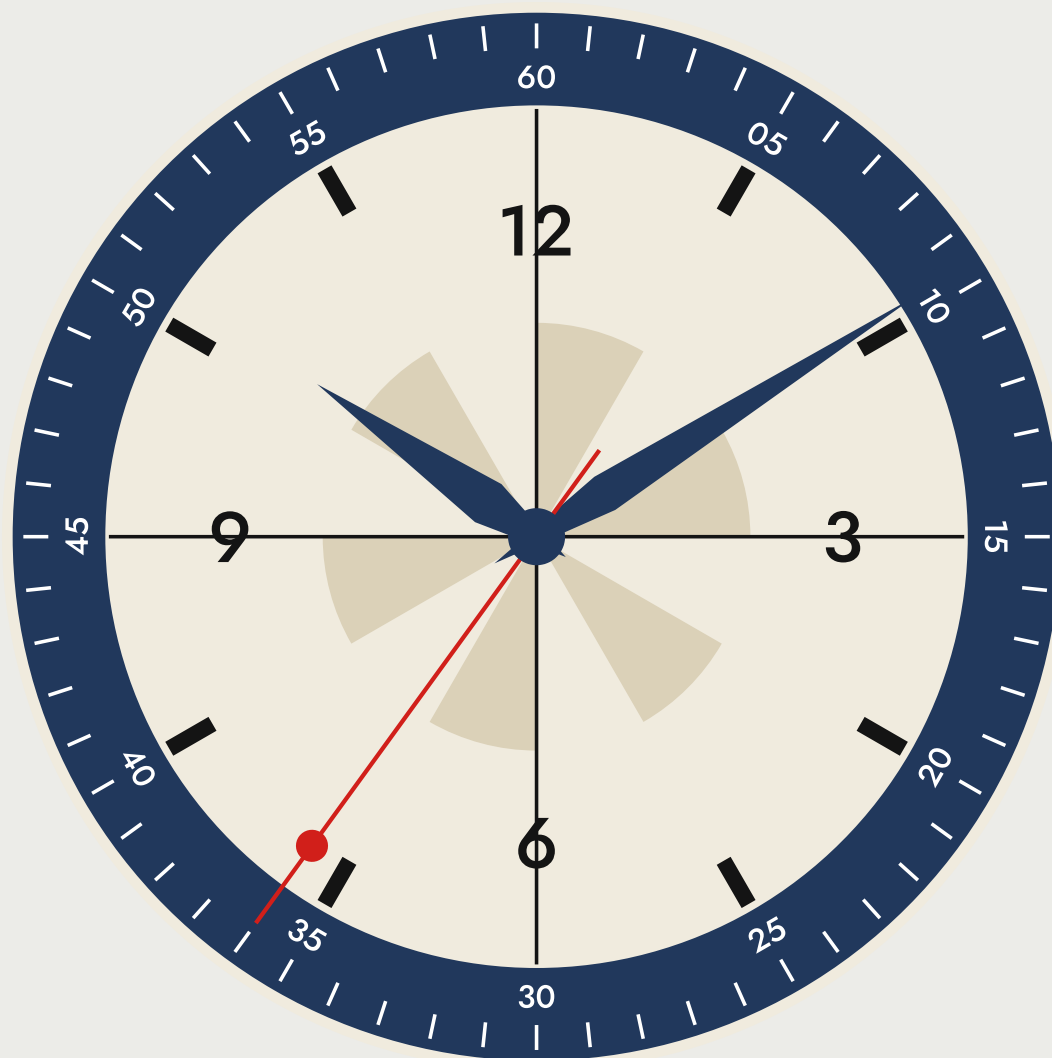
After the German wall-clock school of the 1950s–60s: hairline marks, small numerals, thin hands.



Look for
Line weights carry it: 0.14 for the hours, 0.09 for the minutes, hands under half a millimetre.

Rings top first, then hands. Only what differs from a new ring.

Hours numerals	Radius 10.9 · Size 1.75 · Font Futura Medium ●
Hour lines markers	Radius 14.6 · Length 2.4 · Width 0.14 ●
Minutes ticks	Radius 14.6 · Count 60 · Length 1.3 · Width 0.09 · Skip where markers above ●
Hour hand	Shape baton · Length 8.6 · Width 0.46 · Tail 1.6 ●
Minute hand	Shape baton · Length 13.4 · Width 0.36 · Tail 2 ●
Seconds hand	Shape baton · Length 14 · Width 0.1 · Tail 3.6 ●
Cap	Ø 0.9 ●
Preset	<i>presets/bauhaus.json</i> · Dial 31 mm ○



12.8

Sector dial

After the 1930s scientific dials: a chapter ring, a crosshair, zones of colour.



Look for

Bands do the colour: a chapter ring 2.6 wide, and pies made by Band width = Radius with Every 2. The crosshair is a ticks ring, Count 4, Length 12.

Rings top first, then hands. Only what differs from a new ring.

Hours numerals	Radius 8.6 · Every 3 · Size 2 · Font Futura Medium ●
Hour bars markers	Radius 11.9 · Every 3 · Skip those instead · Length 1.4 · Width 0.45 ●
Minutes ticks	Radius 14.4 · Count 60 · Length 0.7 · Width 0.1 ○
Minute numerals	Radius 12.9 · Start 30° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Placement radial, auto-flip · Size 0.9 · Font Futura Medium ○
Chapter band	Radius 14.7 · Band width 2.6 ●
Crosshair ticks	Radius 12 · Count 4 · Length 12 · Width 0.1 ●
Sectors band	Radius 6 · Count 12 · Every 2 · Band width 6 ○
Hour hand	Shape dauphine · Length 7.5 · Width 1.3 · Tail 1 · at 22 % ●
Minute hand	Shape dauphine · Length 12.6 · Width 1.1 · Tail 1.4 · at 18 % ●
Seconds hand	Shape lollipop · Length 13.4 · Width 0.12 · Tail 3 · Head · ring · disc 0.9 · at 80 % ●
Cap	Ø 1.6 ●
Preset	<i>presets/sector.json</i> · Dial 30 mm ○



12.9

Gauge

A 270° scale with a red zone: positions over a partial span mark both ends.



Look for

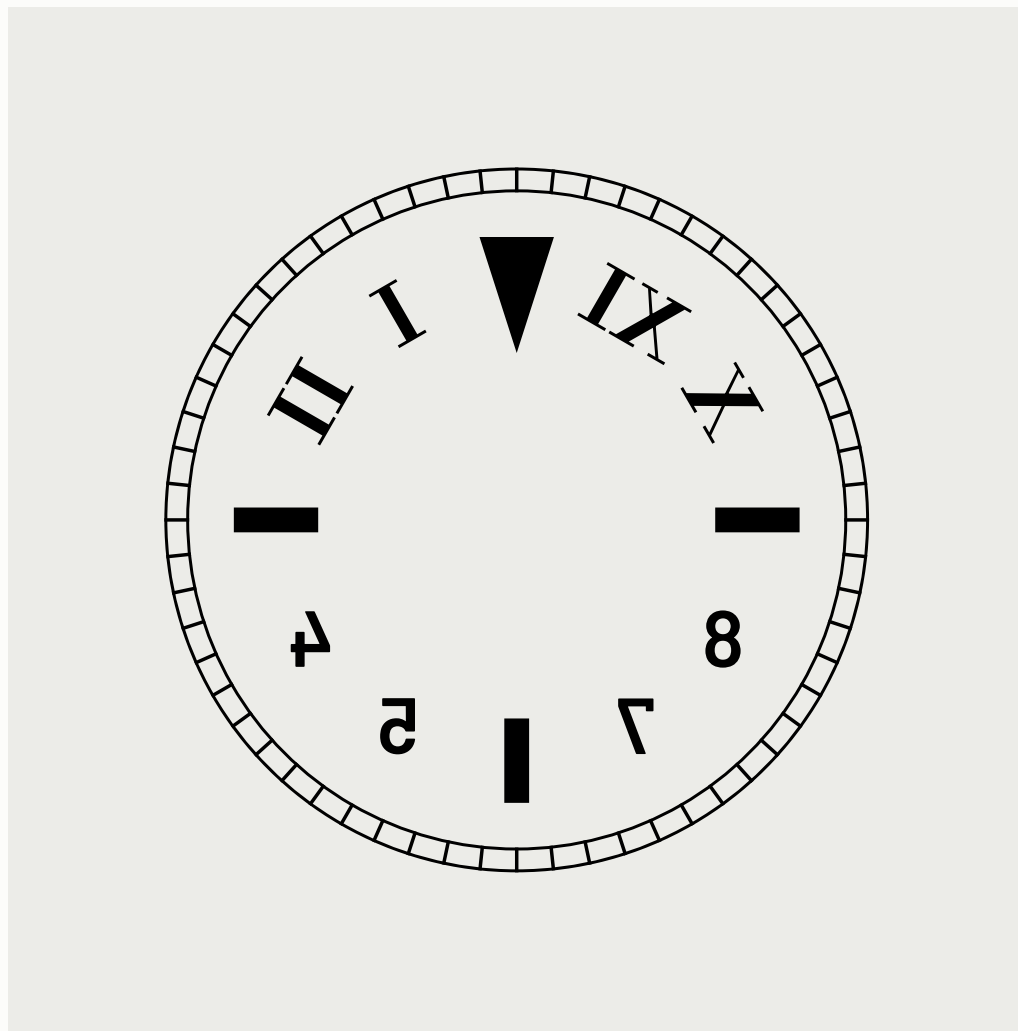
Start -135°, Span 270°. A partial span marks both ends, so Count 10 gives 0 to 90. The red zone **knocks out** everything above it with a 0.12 clearance.

Rings top first, then hands. Only what differs from a new ring.

Scale numerals	Radius 11.3 · Count 10 · Start -135° · Span 270° · Labels Numbers · First number 0 · Step 10 · Placement radial, auto-flip · Size 1.5 · Font Helvetica Neue Medium ●
Scale marks markers	Radius 14.3 · Count 10 · Start -135° · Span 270° · Shape wedge · Length 1.6 · Width 0.34 · Inner width × 0.3 ●
Scale ticks ticks	Radius 14.3 · Count 46 · Start -135° · Span 270° · Length 0.8 · Width 0.1 · Skip where markers above ●
Rail band	Radius 14.5 · Count 1 · Start -135° · Span 270° · Band width 0.12 ●
Red zone band	Radius 14.5 · Count 1 · Start 108° · Span 27° · Band width 0.9 · Knock out everything above · Clearance 0.12 ●
Minute hand	Shape arrow · Length 12.6 · Width 0.4 · Tail 2.6 · Tip length 2.2 · Head · ring · disc 1.5 ●
Cap	Ø 1.2 ●
Preset	<i>presets/gauge.json</i> · Dial 30 mm ○

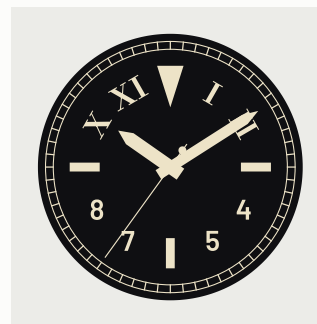
Files in and out

Vectors at true size, a one-colour mask for production, and your own shapes coming in.



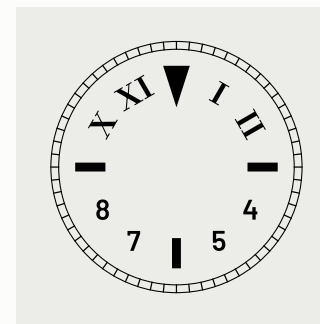
Watch

This is the production mask, mirrored: what goes on the film for toner transfer. Check its thinnest line at 100 % before you etch: a 0.09 mm tick is two pixels at 600 dpi.



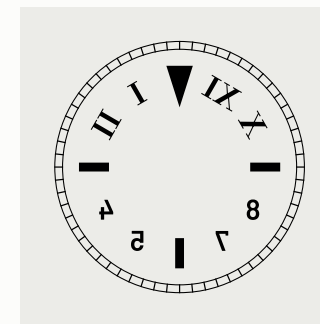
design

PDF · SVG · PNG



production

one colour, no plate, no hands



production + mirror

for toner transfer



Export

☒ Production: mono print mask

☒ Mirror: toner transfer

☐ Live text in PDF · SVG (fonts n...

PNG dpi



PDF · SVG · PNG

True size.

A 30 mm dial opens 30 mm wide. PNG at your dpi: 30 mm at 600 dpi is 709 pixels.



Live text

Off by default.

Keeps numerals editable. The fonts are then needed where the file is opened, and on-path text becomes one object per letter.



In: SVG · PDF · AI

Your own shapes.

Draw pointing up; the artboard's centre is the anchor. Black is ink, white cuts holes. Outline text first.

Recipes and help

Short constructions to build on, each with a dial that uses it. Then what to check when something looks wrong.



Track

Railroad

Ticks, Count 60. Two bands 0.12 wide: one at the ticks' Radius, one at Radius – tick Length + 0.12. Page 19.



Track

Fifths of a second

Ticks Count 60 above ticks Count 300. The 300 Skip where: all marks above.



Hours

24-hour ring

Numerals, Labels Custom: 24, 13, 14 ... 23, at a smaller Radius and Size. Page 15.



Hours

Two fonts, one circle

Two numerals rings on one Radius; each Custom list leaves the other's positions empty. Page 19.



Minutes

05 to 60

Numerals, Labels Numbers, First number 5, Step 5, Digits 2, Start 30°. Page 22.



Text

A wordmark

Numerals, Count 1, Start 0° or 180°, Labels Custom, Placement on path. Page 17.



Colour

A gap round numerals

A band below the numerals; Knock out: markers & numerals above; Clearance 0.3.



Colour

Sector dial

A band, Count 12, Every 2, Band width equal to Radius. Page 22.



A ring is gone

Look at its checkbox in the list, its Every and Offset, and whether it skips positions a ring above takes.



No numeral at 3

The date window took it: markers and numerals leave its position empty.



Slow

The log should say “shape combining: skia-pathops”. If not, install it: page 2.



Start over

Export → Settings → Defaults. The last session is in *~/Library/Application Support/DialTool*.



When something is wrong

Read the log, compare the previews, send both.

- 1 Open Export and read the log. At launch it says “preview: canvas” and “all controls connected”.
- 2 If the preview looks wrong, switch Canvas to PDF and compare.
- 3 Send a screenshot and the log. If the window doesn't build at all, run *tools/dial_probe.py* in DrawBot and send its report.



Where things live

One repository.

Code and docs	github.com/duztman/dial-tool
Words	<i>docs/glossary.md</i>
Presets	<i>docs/guide/presets</i>
This guide	<i>docs/guide/guide.html</i> , built by <i>tools/make_guide.py</i> ; its rules are in <i>docs/design.md</i>